

GA

Genetic Algorithms



A **genetic algorithm** (or **GA**) is a search technique used in computing to find true or approximate solutions to optimization and search problems.

Example for GA



Meet Nixie, a 3d printed
quadruped robot

Evolution

Evolution is the process by which modern organisms have descended from ancient ones



Microevolution

Microevolution is evolution within a single population; (a population is a group of organisms that share the same gene pool). Often this kind of evolution is looked upon as change in gene frequency within a population

For evolution to occur

Heredity

Information needs to be passed on from one generation to the next

Genetic Variation

There has to be differences in the characteristics of individuals in order for change to occur

Heredity

Heredity is the transfer of characteristics (or traits) from parent to offspring through genes



Genetic Variation

Is about variety in the population and hence presence of genetic variation improves chances of coming up with “something new”

The primary mechanisms of achieving genetic variation are:

Mutations

Gene Flow

Reproduction

Mutation

It is a random change in DNA

It can be beneficial, neutral or harmful to the organism

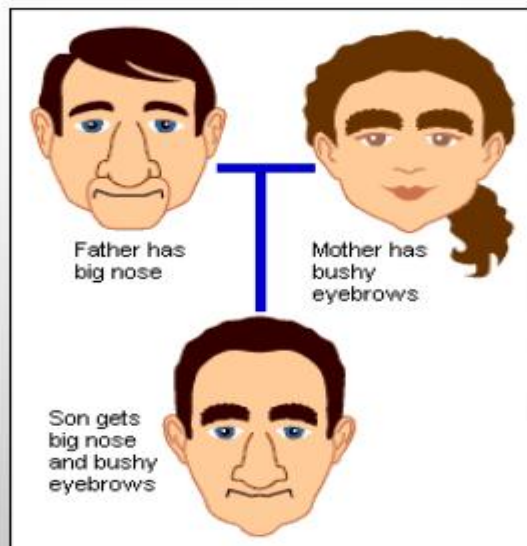
Gene Flow

Migration of genes from one population to another

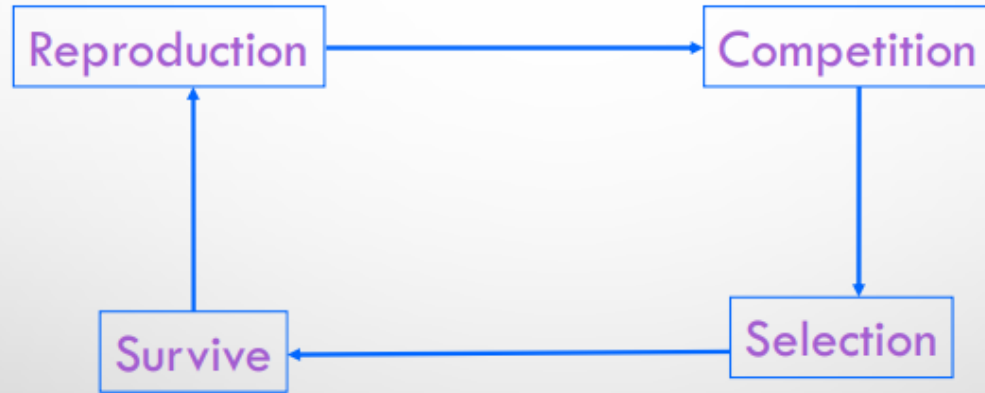
If the migrating genes did not exist previously in the incident population then such a migration adds to the gene pool

Reproduction

This type of producing young can introduce new gene combinations through genetic shuffling

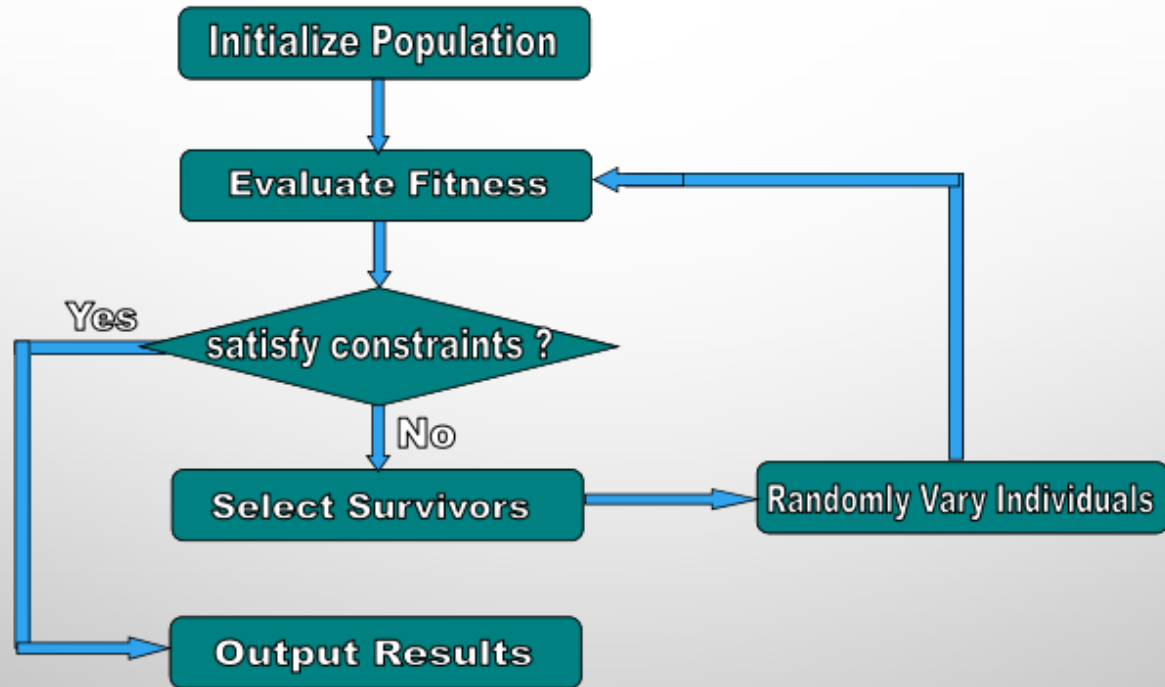


Based On Darwinian Paradigm



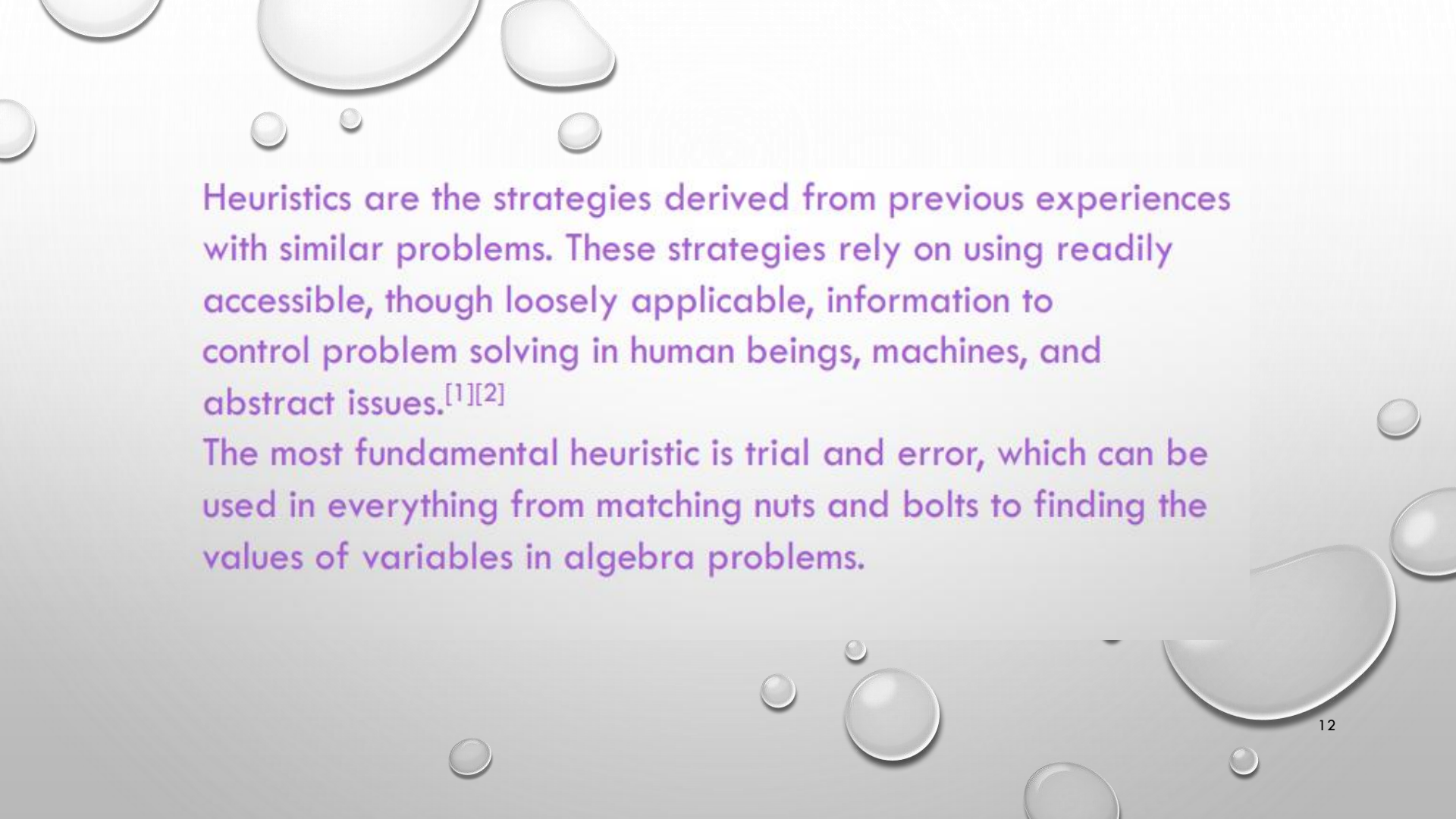
principles first laid down by Charles Darwin, “survival of the fittest,” because in nature, competition among individuals for scanty resources results in the fittest individuals dominating over the weaker ones.

Conceptual Algorithm



What is GA ?

- A **genetic algorithm** (or **GA**) is a search technique used in computing to find true or approximate solutions to optimization and search problems.
- Genetic algorithms are categorized as global search **heuristics**.
- Genetic algorithms are a particular class of evolutionary algorithms that use techniques inspired by evolutionary biology such as inheritance, mutation, selection, and crossover (also called recombination).

The slide features a light gray background with several realistic water droplets of various sizes scattered across the top and bottom edges. The droplets have highlights and shadows, giving them a three-dimensional appearance.

Heuristics are the strategies derived from previous experiences with similar problems. These strategies rely on using readily accessible, though loosely applicable, information to control problem solving in human beings, machines, and abstract issues.^{[1][2]}

The most fundamental heuristic is trial and error, which can be used in everything from matching nuts and bolts to finding the values of variables in algebra problems.

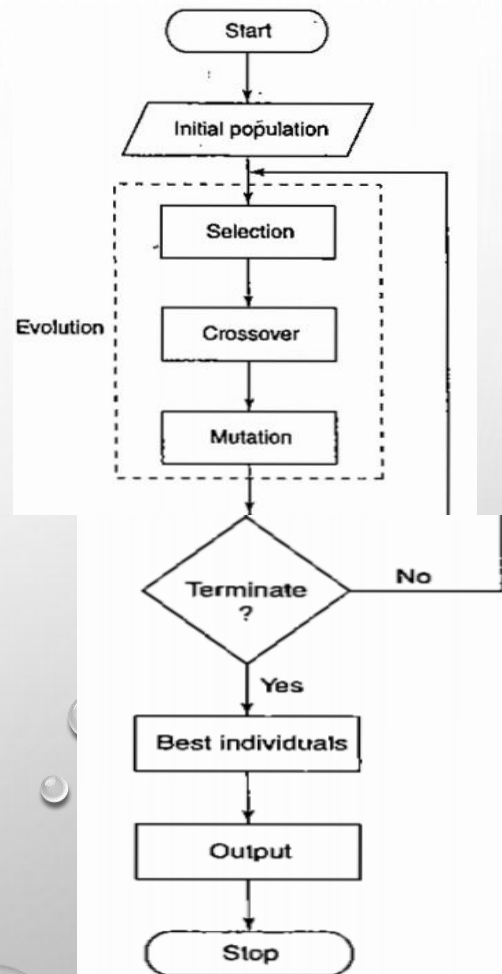
- Genetic algorithms are implemented as a computer simulation in which a population of abstract representations (called chromosomes or the genotype or the genome) of candidate solutions (called individuals, creatures, or phenotypes) to an optimization problem evolves toward better solutions.
- Traditionally, solutions are represented in binary as strings of 0s and 1s, but other encodings are also possible.

Step 1: Create a random initial state: An initial population is created from a random selection of solutions (which are analogous to chromosomes). This is unlike the situation for symbolic AI systems, where the initial state in a problem is already given.

Step 2: Evaluate fitness: A value for fitness is assigned to each solution (chromosome) depending on how close it actually is to solving the problem (thus arriving to the answer of the desired problem). (These “solutions” are not to be confused with “answers” to the problem; think of them as possible characteristics that the system would employ in order to reach the answer.)

Step 3: Reproduce (and children mutate): Those chromosomes with a higher fitness value are more likely to reproduce offspring (which can mutate after reproduction). The offspring is a product of the father and mother, whose composition consists of a combination of genes from the two (this process is known as “crossover”).

Step 4: Next generation: If the new generation contains a solution that produces an output that is close enough or equal to the desired answer then the problem has been solved. If this is not the case, then the new generation will go through the same process as their parents did. This will continue until a solution is reached.

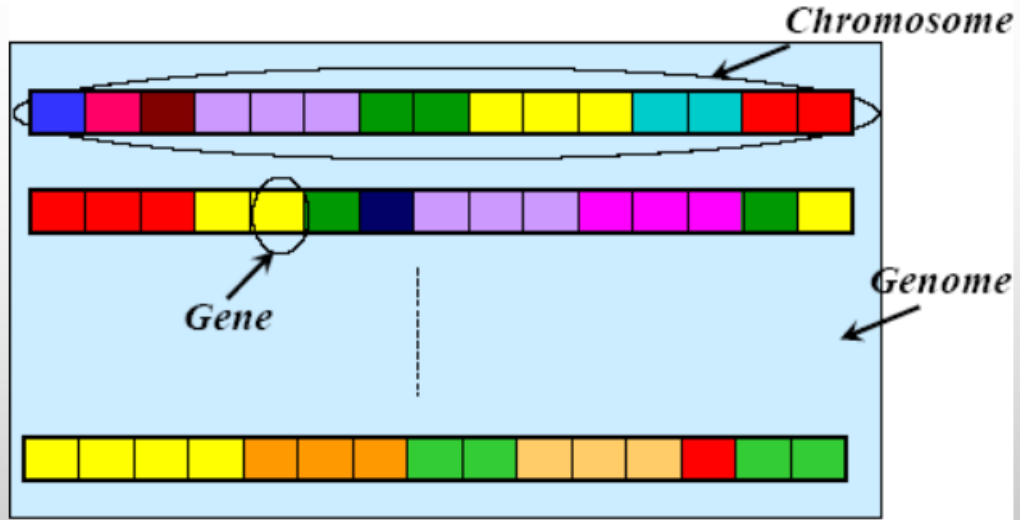


Basics of GA

- The most common type of genetic algorithm works like this:
 - ☐ A population is created with a group of individuals created randomly.
 - ☐ The individuals in the population are then evaluated.
 - ☐ The evaluation function is provided by the programmer and gives the individuals a score based on how well they perform at the given task.
 - ☐ Two individuals are then selected based on their fitness, the higher the fitness, the higher the chance of being selected.
 - ☐ These individuals then "reproduce" to create one or more offspring, after which the offspring are mutated randomly.
 - ☐ This continues until a suitable solution has been found or a certain number of generations have passed, depending on the needs of the programmer.

- The evolution usually starts from a population of randomly generated individuals and happens in generations.
- In each generation, the fitness of every individual in the population is evaluated, multiple individuals are selected from the current population (based on their fitness), and modified (recombined and possibly mutated) to form a new population.

- The new population is then used in the next iteration of the algorithm.
- Commonly, the algorithm terminates when either a maximum number of generations has been produced, or a satisfactory fitness level has been reached for the population.
- If the algorithm has terminated due to a maximum number of generations, a satisfactory solution may or may not have been reached.



- **Genome** - collection of all chromosomes for an individual

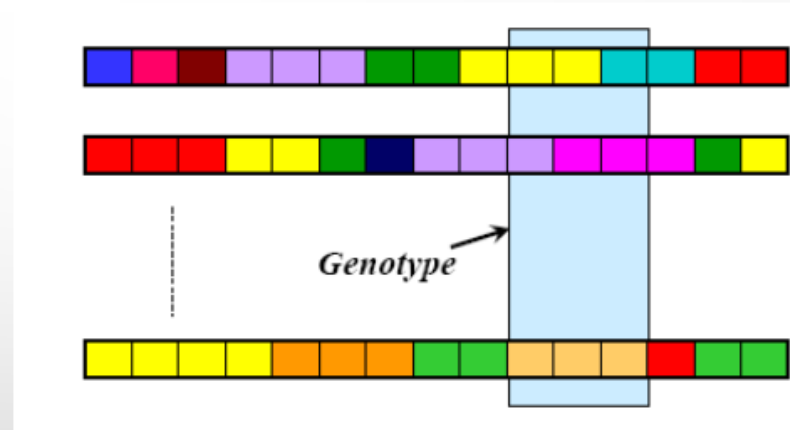
Genotype and phenotype

- **Genotype:**

- Particular set of genes in a genome

- **Phenotype:**

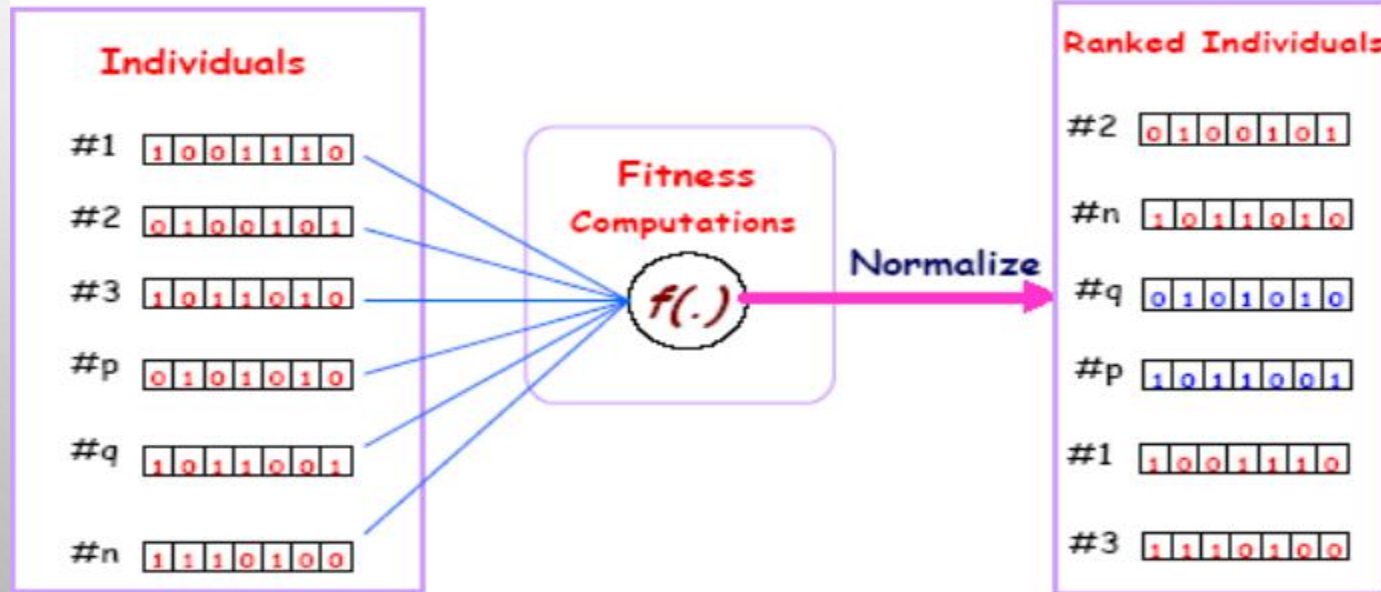
- Physical characteristic of the genotype (smart, beautiful, healthy, etc.)



GA requirements

- The fitness function is defined over the genetic representation and measures the *quality* of the represented solution.
- The fitness function is always problem dependent.

A fitness function



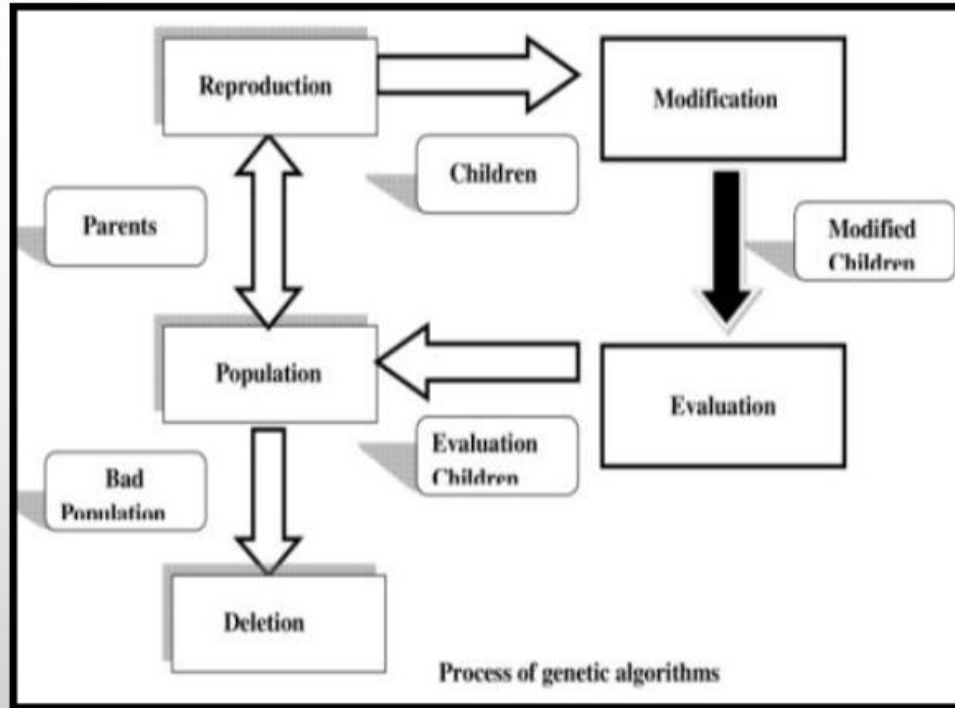
GENERAL ALGORITHM FOR GA

- **Initialization**
- Initially many individual solutions are randomly generated to form an initial population. The population size depends on the nature of the problem, but typically contains several hundreds or thousands of possible solutions.
- Traditionally, the population is generated randomly, covering the entire range of possible solutions (the *search space*).
- Occasionally, the solutions may be "seeded" in areas where optimal solutions are likely to be found.

General algorithm for GA

- **Selection**

- During each successive generation, a proportion of the existing population is selected to breed a new generation.
- Individual solutions are selected through a *fitness-based* process, where fitter solutions (as measured by a fitness function) are typically more likely to be selected.
- Certain selection methods rate the fitness of each solution and preferentially select the best solutions. Other methods rate only a random sample of the population, as this process may be very time-consuming.
- Most functions are stochastic and designed so that a small proportion of less fit solutions are selected. This helps keep the diversity of the population large, preventing premature convergence on poor solutions. Popular and well-studied selection methods include roulette wheel selection and tournament selection.



Representation

Chromosomes could be:

- Bit strings (0101 ... 1100)
- Real numbers (43.2 -33.1 ... 0.0 89.2)
- Permutations of element (e11 e3 e7 ... e1 e15)
- Lists of rules (R1 R2 R3 ... R22 R23)
- Program elements (genetic programming)
- ... Any data structure ...

GA requirements

- A typical genetic algorithm requires two things to be defined:
 - ✓ A genetic representation of the solution domain, and
 - ✓ A fitness function to evaluate the solution domain.
- A standard representation of the solution is as an array of bits. Arrays of other types and structures can be used in essentially the same way.
- The main property that makes these genetic representations convenient is that their parts are easily aligned due to their fixed size, that facilitates simple crossover operation.
- Variable length representations may also be used, but crossover implementation is more complex in this case.
- Tree-like representations are explored in genetic programming.

15.2.2 Chromosomes

All the genetic information gets stored in the chromosomes. Each chromosome is build of deoxyribonucleic acid (DNA). In humans, chromosomes exist in pairs (23 pairs found). The chromosomes are divided into several parts called *genes*. Genes code the properties of species, i.e., the characteristics of an individual. The possibilities of combination of the genes for one property are called *alleles*, and a gene can take different alleles. For example, there is a gene for eye color, and all the different possible alleles are black, brown, blue and green (since no one has red or violet eyes!). The set of all possible alleles present in a particular population forms a *gene pool*. This gene pool can determine all the different possible variations for the future generations. The size of the gene pool helps in determining the diversity of the individuals in the population. The set of all the genes of a specific species is called *genome*. Each and every gene has a unique position on the genome called